
BFMC 2025 : Status Report 4**Date :** Monday, April 21, 2025**Prepared by :** Pothole Avoidance**Reporting Period :** Tuesday, March 17, 2025 - Monday, April 21, 2025

Overview of project status

The past month was focused on in depth improvements. This was a sharp contrast from the approach that had been taken in the months of February and March. In those months, a working vehicle was essential to make sure we were able to qualify for the finals. This led to us finding quick fixes and temporary solutions to some problems we encountered. The qualifications over, it was time to tear down those quick fixes and get at the root of the problem. This meant that time was dedicated to adding an encoder for more precise odometry, extensive research for the design of a good power board, finishing up the GUI and designing a pipeline for training our object detection.

Current Phase

The qualifications identified some key shortcomings that were critical to the reliability of our project. As was outlined in our additional status report for the qualifications¹, the major issue we identified was the unreliability of the speed commands sent. Our control being heavily dependent on these commands, we needed to address this at its core. Two separate tasks were therefore dedicated to solving this issue : adding an encoder and designing a new power board.

Given that the next big performance is in June, the focus of the month of April was on developing in depth solutions, taking the time for research, prototyping and testing knowing that this time was available to us. The critical dependencies of the project lay in the hands of the hardware team : the software team is waiting on the hardware to begin testing. Software is in an advanced state and is mostly focused on fine tuning and fixing small non-critical bugs through simulation testing.

Task Status - Overview

The end of the semester meant it was more difficult for some team members to find time to dedicate to their tasks. However, given the current phase of the project being one of more in depth research and development with less pressure and time constraints, this was not too much of an issue. Now that finals are coming to an end, more time can be dedicated to wrapping up these tasks before entering the integration phase. As usual, the full status of the tasks can be found in the task tracker table export², with the main concerned tasks being described in more detail in the following.

Encoder Addition : This task is of high priority and is close to completion. Its undertaking is dependent on the intersection of both hardware and micro controller design. The micro controller design portion is responsible for receiving the appropriate signal from the encoder, processing and filtering the data before the derivation of the velocity. This is all done in parallel to the IMU data filtering and controlling of both the motors and servos. On the hardware side, research into which magnets need to be used for optimal sensor function, the design of a mount on which to install the encoder and magnet as well as connections to the rotary shaft are required. All this research and installation is the first step before calibration, testing and implementation which are due at the end of April.

Power Distribution Board Design : This task is of high priority but is not yet complete. The design of a power board was already something that had been done over the months of December and January. However, the power board we had designed did not include voltage regulation as we did not think it was such a critical issue. This task is difficult precisely because it includes the design of a high precision voltage regulator on the power distribution board. This requires us to take into account the

¹See *status_report_qualifications.pdf*, available on our GitHub

²See *task_tracker_export_status_4*

characteristics of the motor that we are using, as well as the ideal voltage and current range that we output to. The intricate nature of the design makes it such that more time is required for this design. On top of consulting the various data sheets available, tools like TI power designer and Altium are being used for a design of the highest quality.

Object Detection Pipeline : The idea behind creating an object detection pipeline comes from the necessity to be able to both evaluate and improve our object detection algorithms. The process of adding new frames obtained from the camera or from other footage, labeling them and keeping the right balance of signs/vehicles/pedestrians can be both confusing and a time sink if not well structured. Therefore, the design of this pipeline to add new images and evaluate how it influences the models we are training greatly improves our ability to create a reliable and accurate object detection model. A detailed outline of this pipeline can be found in the supporting documentation³.

Blocking Points and Encountered Issues

Encoder Placement & Communication : The first challenge to placing the encoder on the car was to be able to get it working and receive values. This proved to be a challenge because while we initially wanted to connect to the SPI pins on the sensor board, these proved to be too small for our team's soldering capabilities. We therefore defaulted to using the PWM output pins. However, this required extra work on the micro controller's side to measure the duty cycle and input pin interrupts.

Another more challenging problem that was encountered was the placement of the magnet and the sensor on the car. Our initial idea was to glue the magnet on the motor and place the encoder next to it. However, results showed that even the slight misalignment on the order of a millimeter or two made the readings from the encoder noisy and difficult to work with. We are thus currently working on connecting another gear next to the driveshaft on which we can precisely mount both the magnet and the encoder. This will result in a clean magnetic field with which we can work.

Power Distribution Uncertainties : One of the important requirements in the power distribution board design is the nominal voltage and current for which we are designing the board. Our initial idea was to use an off-the-shelf voltage regulator board, that we would connect in series between the motor and the power board. This did not work because of the lack of precision of such off-the-shelf voltage regulators. We need a very small regulation ($\sim 8.0V$ to $7.4V$) for a high current. The pre-built voltage converters are good for a large range of voltages, but not for precise conversions.

Creating a custom voltage regulating power board requires that we know the specifications of the devices we are working with : the motor in particular. Unfortunately, its data sheet does not specify well the nominal current, and even when we tried to contact the manufacturer they were unable to give us a clear relationship between current and voltage : « The current draw on the system will depend on the load.[...] There is no direct correlation between the load and amp draw, there are many other factors that will play in to the current the system will draw. » We were disappointed by the response for the manufacturer and will have to rely on our best effort guess in the design of the board.

Next Steps

The month of May will see us transition to integration testing with these new features. This is essential as the finals are approaching in less than two months, and we want to have a reliable vehicle by the beginning of June (*i.e.* given the same initial conditions, be able to get the same run 10 times out of 10). This integration testing is crucial as some problems are currently undetectable because they are covered by more important problems. As the month of May coincides with the end of the school year, some members will have more time on their hands to dedicate to the project. Other members are starting or continuing work in industry, and will have to adapt their schedules. Through this, the plan is to build more track portions to test on (roundabout, highway, *etc.*) which will enable us to evaluate the performance of the car in the real world all while tuning the newly added features.

³See *software_pdr_march_23.pdf*